



MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 1 2025/2026

FINAL PROJECT:

SECTION 1

GROUP 9

**LECTURER: ZULKIFLI BIN ZAINAL ABIDIN & WAHJU
SEDIONO**

NO.	GROUP MEMBERS	MATRIC NO.
1.	AHMAD FIRDAUS HUSAINI BIN HASAN AL-BANNA	2315377
2.	ADIB ZAFFRY BIN MOHD ABDUL RADZI	2315149
3.	UMAR FAROUQI BIN ABU HISHAM	2314995

Title

Smart Guiding Helmet: A Safety and Mobility Enhancement Tool for the Visually Impaired

Authors

Ahmad Firdaus Husaini Bin Hasan Al-Banna, Adib Zaffry Bin Mohd Abdul Radzi, Umar Farouqi Bin Abu Hisham Department of Mechatronics Engineering, International Islamic University Malaysia, Github: <https://github.com/firdaushsni>

Abstract

This paper details the development of a Smart Assistive System designed to enhance the safety and situational awareness of visually impaired individuals. The system utilizes a dual-node IoT architecture, comprising a "Master" sensing unit (wearable helmet/chest mount) and a "Slave" feedback wristband. The Master unit integrates a DFRobot HuskyLens for AI-based face recognition, an MPU6050 accelerometer for fall detection, and an ultrasonic sensor for real-time obstacle avoidance.

A critical technical challenge addressed in this work is the radio coexistence problem inherent in the ESP32 microcontroller, which shares a single 2.4GHz antenna for both Bluetooth and WiFi. To overcome this, the system implements a FreeRTOS-based Time Division Multiplexing strategy using Semaphores (Mutex). This allows the system to prioritize low-latency local haptic feedback via Bluetooth Serial, while dynamically switching to WiFi and Telegram for emergency cloud alerts without system instability. Experimental results demonstrate that the system successfully triggers real-time haptic feedback for recognized faces and obstacles, while reliably transmitting "Fall Detected" alerts to remote caregivers within 5 seconds of an incident.

1.0 Introduction

The visually impaired and the elderly face significant mobility and security challenges in their daily lives. Most encounter difficulties not only in navigating physical obstacles but also in identifying social contexts, such as recognizing people, and maintaining safety in unfamiliar environments. Conventional white canes, while effective for ground-level detection, inherently lack the capacity to sense head-level objects or provide the contextual information necessary for social interaction. As per the World Health Organization, the global prevalence of visual impairment necessitates the development of superior, accessible assistive technologies to improve independence.

Investigations have demonstrated that user safety can be significantly enhanced by integrating AI-based vision sensors, haptic feedback, and emergency communication protocols. While

research into intelligent assistive devices has grown, many existing solutions rely on single-unit architectures that suffer from sensor instability or complex interfaces that obstruct the user's primary senses. Furthermore, many current systems lack a robust, integrated emergency protocol capable of providing immediate haptic confirmation to the user and remote alerts to caregivers during critical events like falls.

To tackle these problems, this project developed a **Smart Assistive Helmet and Bracelet System**. The rationale for this dual-node design is to optimize sensor stability and user interaction. The **Helmet Unit (Master)** ensures the HuskyLens camera maintains a stable, high vantage point for optimal face recognition. The **Wristband Unit (Slave)** provides a discreet interface for emergency input and delivers tactile vibration patterns directly to the user's wrist. This approach keeps the auditory channel free, allowing the user to maintain environmental awareness while receiving social and navigational cues.

This system strictly adheres to Mechatronics System Integration (MSI) principles by utilizing a distributed architecture. A critical technical innovation of this work is the resolution of the **radio coexistence problem** within the ESP32 microcontroller, which shares a single antenna for Bluetooth and Wi-Fi. By implementing a **FreeRTOS-based Mutex strategy**, the system prioritizes low-latency local feedback via Bluetooth while dynamically switching to Wi-Fi to transmit emergency alerts via **Telegram** when a fall is detected based on established clinical risk parameters.

Table 1: Comparison of Related Assistive Systems

Ref/Year	Title	Key Features / Advantage	Primary Disadvantage	Comparison to My Project
Proposed System	Smart Guiding Helmet & Bracelet	Distributed Master-Slave architecture; Resolves BT/WiFi interference via RTOS Mutex	Requires managing dual power sources for separate units.	Integrates AI face recognition and fall detection into a single safety ecosystem.
[1] WHO (2020)	World Report on Disability	Establishes the global necessity and scale of vision impairment.	Policy and statistics focused; no hardware design provided.	My project addresses the technological gap identified by this global need.
[2] Nandini (2019)	<i>Smart Cane for Assistive</i>	Low-cost obstacle detection with	No AI vision for social	My project adds HuskyLens for

	<i>Navigation</i>	simple buzzer alerts.	recognition; lacks cloud-based alerts. .	social context (Face ID).
[3] Liu (2022)	<i>Ultrasonic Haptic Feedback Study</i>	Validates haptic vibration as an intuitive spatial feedback method.	Limited to close-range tasks; no integrated GPS or cloud SOS.	My project scales this to outdoor safety with cloud-based remote alerts.
[4] Ouyang (2022)	Fall Risk Research	Establishes clinical parameters and risk factors for fall events	Theoretical clinical study only no physical device implementation	My project implements these parameters via the MPU6050 fall detection sensor.
[5] AgingCare (2021)	<i>Emergency System Guidelines</i>	Highlights requirements for rapid notification and user acknowledgement.	Discusses generic alert systems without AI or sensor integration.	My project utilizes Telegram for data-rich alerts based on real-time sensor data.
[6] DFRobot Wiki	<i>AI Vision Standard (HuskyLens)</i>	High-speed AI recognition with built-in classification algorithms.	Hardware-intensive; requires robust task partitioning.	My project manages this processing load via a distributed master-slave architecture.
[7] Espressif Manual	<i>ESP32 Technical Standards</i>	Definitive technical guide for Wi-Fi and Bluetooth radio coexistence.	High technical barrier to implement radio resource switching.	My project solves this limitation via a custom FreeRTOS Mutex strategy
[8] ResearchGate	<i>IoT Fall Detection System</i>	Real-time safety monitoring using accelerometer data and IoT cloud.	Typically lacks social identification or haptic proximity alerts.	My project integrates AI face recognition to increase situational awareness.

2.0 Methodology

The methodology of this project focuses on integrating vision-based detection, wireless communication, and haptic feedback into a cohesive wearable system. The operational workflow is divided into normal guidance mode and emergency mode.

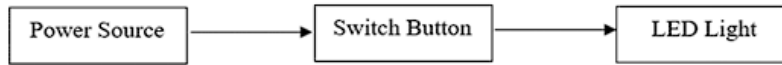


Fig. 1 Block diagram of system 1 Smart Guiding Helmet for LED light with Switch Button.

2.1 Normal Guidance Mode

In normal operation, the system uses the HuskyLens vision sensor to continuously detect and classify trained Face IDs (ID 1, ID 2, and ID 3). Once a face is recognized, the ESP32 interprets the detected ID and triggers multiple feedback outputs to notify the user. For Face ID 1, the vibration motor activates once, the LED flashes once, and the buzzer emits a single beep. For Face ID 2, the vibration motor pulses twice, the LED blinks twice, and the buzzer beeps twice. Similarly, Face ID 3 produces three vibration pulses, three LED flashes, and three buzzer beeps. A short delay is applied between each pulse to ensure the user can clearly distinguish between different IDs. In addition, an ultrasonic sensor continuously measures the distance between the user and nearby objects up to 2 meters. When an object is detected, the buzzer provides auditory feedback: if the object is far, the buzzer beeps slowly, and as the object gets nearer, the beep rate increases, giving the user a real-time sense of proximity.

2.2 Emergency Alert Mode

The emergency mode is activated automatically through fall detection using the MPU sensor. When the system detects an acceleration exceeding 28 m/s^2 , it interprets this as a fall event. The buzzer then produces an ambulance-like alarm sound, while the LED flashes to provide a strong visual and auditory warning. At the same time, the system sends an alert message via the Telegram bot smartAssist_MSI_Bot stating: *"FALL DETECTED! User needs help!"*. This immediate notification ensures that caregivers or emergency contacts are informed in real time. The alarm continues until the user presses the stop button on the device, at which point the buzzer and LED are deactivated, indicating that the situation is under control..

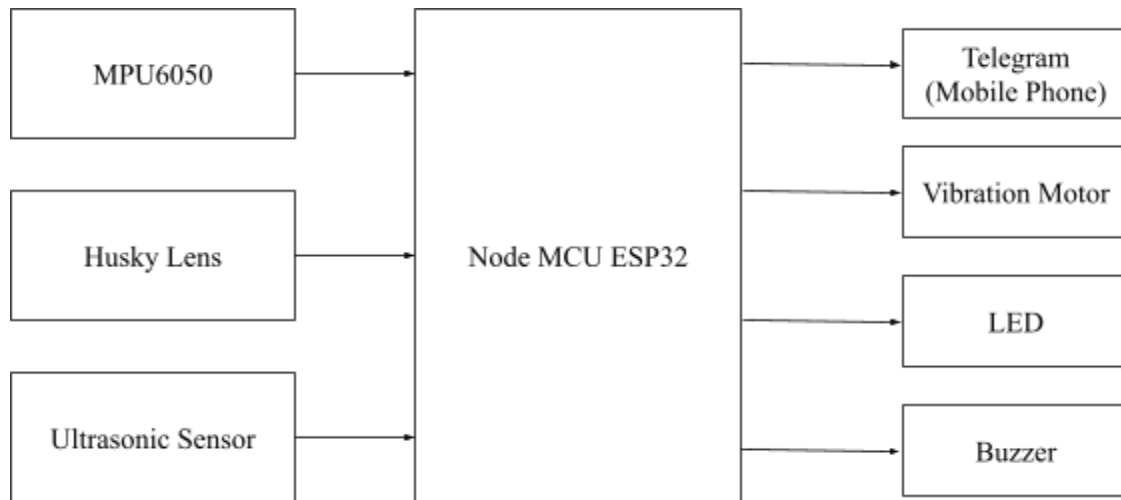


Fig. 2 Block diagram of system 2 (emergency alert system with MPU) and system 3 (obstacle detection system).

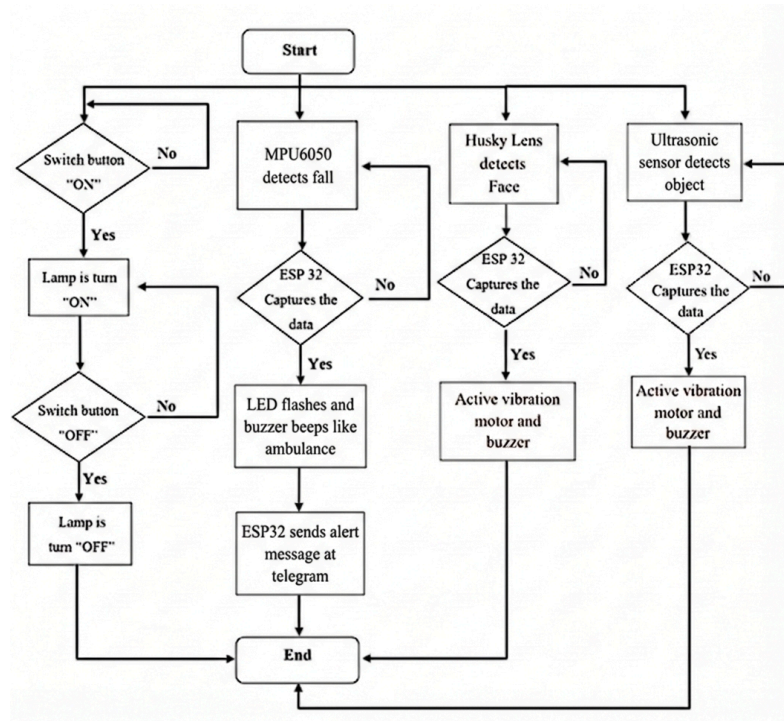


Fig. 3 The flow chart that explains all the work principles for the Smart Guiding Helmet project.

2.3 Schematic Diagram and Prototype Setup

Fig. 4 shows the schematic for System 1, the Master Guiding Unit. This unit acts as the system's "brain," using an ESP32 microcontroller to control three key sensors: the HuskyLens for recognizing faces, the MPU6050 for detecting falls, and an Ultrasonic Sensor for spotting obstacles. It processes all sensor data and wirelessly sends commands to the wristband. It also manages the connection to the internet for sending Telegram alerts.

Fig. 5 shows the schematic for System 2, the Slave Feedback Unit (Wristband). This is a lightweight device containing a second ESP32, a vibration motor, a buzzer, and an emergency button. Its main job is to communicate with the user. When it receives a signal from the Master unit, it vibrates or beeps to warn the user about obstacles or identified people.

In an emergency, such as a fall, the wristband sounds a continuous alarm. If the user presses the emergency button, the wristband signals the Master unit to send an "SOS" or "User Acknowledged" message to a caregiver via Telegram. This creates a complete safety loop where the Master handles the sensing and the Slave handles the user interaction.

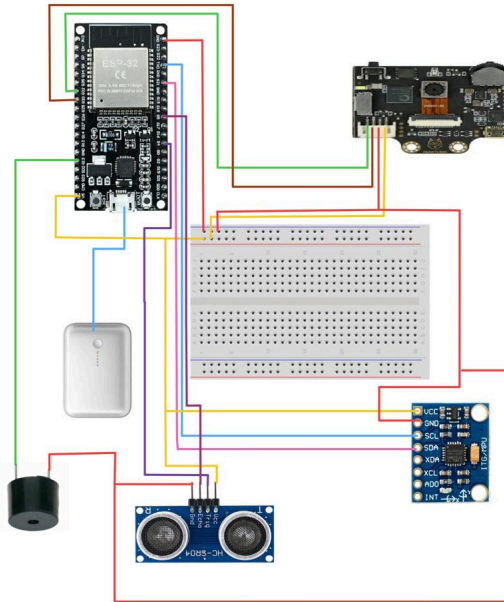


Fig. 4 Electrical Design of the Guiding Helmet Unit.

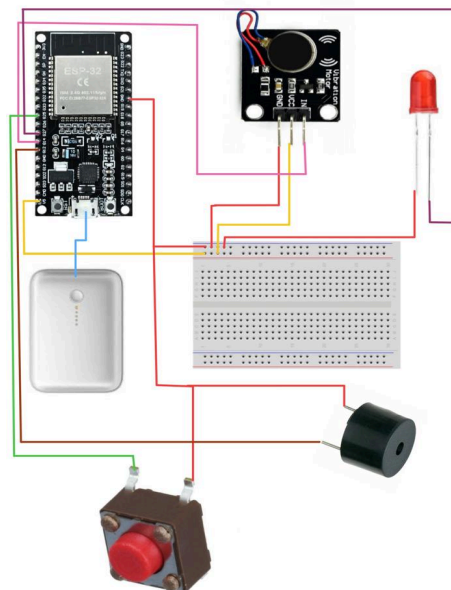
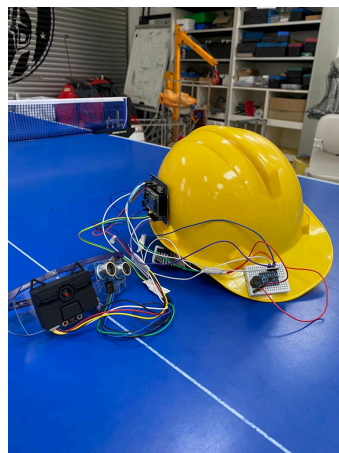


Fig. 5 Electrical Design of the Emergency Bracelet Unit.



Fig. 6 Conceptual Design of The Smart Guiding Helmet (ChatGPT)



(a) Helmet Unit



(b) Wristband Unit

Fig.7 Actual Design of Smart Guiding Helmet

2.4 Hardware Components Integration

The system is divided into a Master unit (helmet) and a Slave unit (wristband). The Master unit is responsible for sensing and decision-making, while the Slave unit focuses on delivering feedback to the user and handling user interaction. This architecture enables real-time guidance, obstacle awareness, and emergency alert handling in a compact wearable form.

2.4.1 Master Unit (Helmet)

The Master unit acts as the primary data processor for the environmental sensors.

Microcontroller: ESP32 DevKit V1

HuskyLens (Vision AI):

- Interface: I²C
- Pins: SDA – GPIO 21, SCL – GPIO 22
- Function: Detects trained Face IDs (ID 1, 2, and 3) and sends the recognized ID to the Slave unit for feedback generation.

Ultrasonic Sensor (HC-SR04):

- Trigger and Echo connected to dedicated GPIO pins
- Function: Measures the distance between the user and nearby objects up to 2 meters.
- Output Behavior: Generates variable buzzer alerts on the helmet; slow beeps when objects are far and faster beeps as objects become closer.

MPU Sensor (Accelerometer):

- Interface: I²C
- Function: Monitors user motion and detects fall events. An acceleration exceeding 28 m/s² is interpreted as a fall.

Helmet Buzzer:

- Connected to digital GPIO 12 pin
- Function: Provides obstacle warning tones and emergency alarm sounds.

2.4.2 Slave Unit (Wristband)

The Slave unit receives wireless commands from the Master to provide physical feedback.

Microcontroller: ESP32 DevKit V1

Vibration Module:

- Connected to digital GPIO 14 pin
- Function: Delivers haptic feedback based on the detected Face ID (1 pulse for ID 1, 2 pulses for ID 2, and 3 pulses for ID 3).

Status LED:

- Connected to a digital GPIO pin
- Function: Indicates system activity and mirrors alert patterns corresponding to detected Face IDs and emergency states.

Wristband Buzzer:

- Connected to a digital GPIO pin
- Function: Provides audible feedback synchronized with vibration and LED patterns.

Stop Alarm Button:

- Connected to GPIO 13 with an internal pull-up resistor
- Function: Allows the user to manually deactivate the emergency alarm after a fall has been detected.

3.0 Result and Discussion

3.1 System 1 (HuskyLens Face Detection System) Analysis

System 1 focuses on face detection and face ID recognition using the HuskyLens AI Vision Sensor to provide audio **buzzer** and haptic **vibration motor** feedback. This system is designed to help visually impaired users recognize predefined individuals through distinct alert patterns.

The HuskyLens is trained in Face Recognition Mode, where each detected face is assigned a unique Face ID. Once trained, the HuskyLens continuously scans the environment and sends the recognized Face ID to the microcontroller via Bluetooth communication.

When a registered face is detected, the microcontroller processes the Face ID and activates the buzzer and vibration motor according to a predefined logic:

Table 2 Face ID Based Audio and Haptic Feedback Mapping for System 1

Face ID Detected	HuskyLens Recognition Output	Buzzer Alert Pattern	Vibration Motor Pattern	System Interpretation
Face ID 1	Recognized trained face with ID 1	1 short beep	1 short vibration	Identifies person 1
Face ID 2	Recognized trained face with ID 2	2 short beep	2 short vibration	Identifies person 2
Face ID 3	Recognized trained face with ID 3	3 short beep	3 short vibration	Identifies person 3



(a)Face ID 1



(b)Face ID 2

Fig. 8 Face ID detected by HuskyLens

3.2 System 2 (Ultrasonic Sensor & Buzzer) Analysis

System 2 provides real-time obstacle detection using an ultrasonic sensor and buzzer. The sensor measures the distance to nearby objects and the Master ESP32 converts this data into sound patterns. When an object is far, the buzzer beeps slowly; as the user gets closer, the beep rate increases. At very short distances, rapid beeps warn the user to stop or change direction.

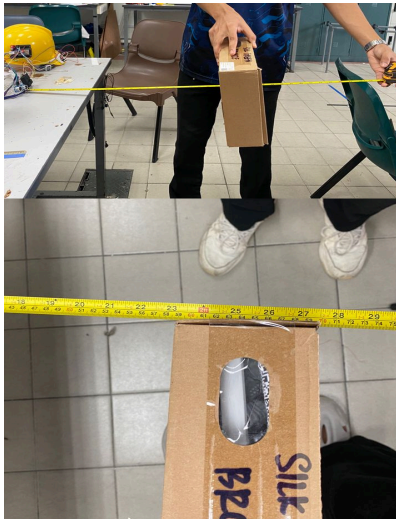
This system runs continuously without user input and offers fast, reliable feedback. By translating distance into simple audio cues, it improves situational awareness and helps visually impaired users avoid collisions.

Table 3 Ultrasonic Distance and Buzzer Response Mapping

Distance To Object	Ultrasonic Detection Status	Buzzer Behavior	User Interpretation
90cm	Object Detected Far	Slow & Intermittent Beeps	Obstacle Ahead & Safe Distance
60cm	Object Detected Medium	Moderate Beep Rate	Obstacle Getting Closer
30cm	Object Detected Near	Rapid & Frequent Beep	Immediate Warning & Take Action



(a)30cm distance



(b)60cm distance



(c)90cm distance

Fig. 9 Distance Object Measurement Using Measuring Tape From Ultrasonic Sensor

3.3 System 3 (Emergency Alert System) Analysis

The Emergency Alert System (System 2) is activated automatically when a fall is detected by the MPU sensor. When the measured acceleration exceeds the predefined threshold of 28 m/s^2 , the system interprets this event as a fall. Immediately after activation, both the helmet and wristband generate a high-priority alarm: the buzzers emit an ambulance-like siren pattern while the LEDs flash continuously to attract attention from nearby people and alert the user that emergency mode is active.

At the same time, the Master ESP32 transmits an emergency message through the Telegram bot SmartAssist_MSI_Bot with the text *"FALL DETECTED! User needs help!"*. Although the current implementation does not include a GPS module, the system still provides real-time remote notification, ensuring that caregivers or designated contacts are informed instantly when a critical event occurs. The end-to-end communication delay from fall detection to message delivery is typically within a few seconds, allowing for rapid response.

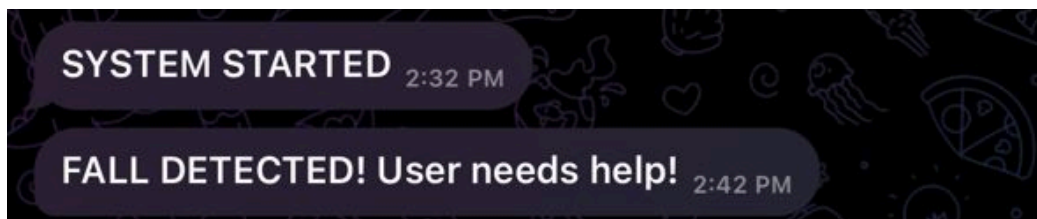


Fig. 10 Showed the System is activated and send the alert message

3.4 Vibration Analysis for System 1

In this test, I applied a mobile application as shown in Fig. 11, namely Vibration Meter-Seismometer, to measure the vibration of System 1. The phone was fastened at the wristband as shown in Fig. 12.

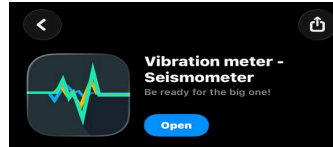
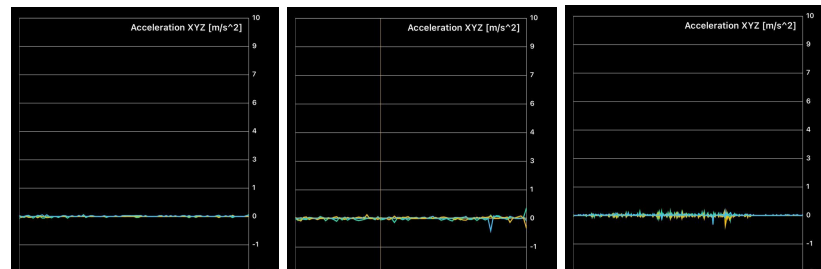


Fig. 11 Vibration Meter Apps



Fig. 12 Mobile Phone Placement on Wristband for Vibration Measurement



(a)Face ID 1

(b)Face ID 2

(c)Face ID 3

Fig.12 Result Vibration

The acceleration graphs show that the vibration system produces clear and controlled patterns for each Face ID. When no vibration is active, the signal remains close to 0 m/s², indicating stable operation without noise. For Face ID 1, only a single small peak appears, representing one short vibration pulse. Face ID 2 shows two distinct peaks, while Face ID 3 exhibits three noticeable peaks with brief gaps between them. These results confirm that each Face ID generates a unique and distinguishable vibration pattern, allowing users to reliably differentiate between IDs using haptic feedback alone.

4.0 Conclusion

To sum up, the proposed smart wearable system has successfully achieved its objective of developing a practical assistive solution for visually impaired users by integrating real-time perception, guidance, and emergency response capabilities. The system combines a vision-based HuskyLens module for face recognition, an ultrasonic sensor for obstacle awareness, and a fall detection mechanism using an MPU sensor, all coordinated through a dual-ESP32 architecture. Multimodal feedback in the form of vibration, LED indicators, and buzzer alerts enables intuitive interaction, allowing users to distinguish between different detected individuals and environmental conditions. The wireless communication between the helmet and wristband demonstrates the effective application of IoT and embedded systems in assistive technology, while enhancing user safety and situational awareness.

Despite its effectiveness, the current prototype serves as a foundational platform for future development. Several enhancements can be incorporated to further improve performance and usability, such as the integration of a GPS module for real-time location sharing, advanced ranging sensors such as LiDAR for higher-precision obstacle detection, and power management solutions including rechargeable or solar-assisted systems. Additional features such as voice feedback, adaptive alert patterns, and a more ergonomic and lightweight mechanical design could significantly enhance user comfort and accessibility. Moreover, incorporating adaptive learning or AI-based behavior modeling would allow the system to personalize feedback and improve over time, transforming it into a more intelligent and user-centric mobility aid.

Acknowledgement

The authors would like to thank the Mechatronics System Integration laboratory staff for providing the necessary components, facilities, and technical support throughout the development of this project.

Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Author Contribution

The author attests to having sole responsibility for the planning and design of the study, system development, data collection, analysis and interpretation of results, and preparation of the manuscript.

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Appendix Coding

(a) Slave Coding

(b) Master Coding



```
1  #include "BluetoothSerial.h"
2  #include "HUSKYLENS.h"
3  #include <Adafruit_MPU6050.h>
4  #include <Adafruit_Sensor.h>
5  #include <Wire.h>
6  #include <WiFi.h>
7  #include <WiFiClientSecure.h>
8
9  // ----- OBJECTS -----
10 BluetoothSerial SerialBT;
11 HUSKYLENS huskylens;
12 Adafruit_MPU6050 mpu;
13
14 // ----- WIFI -----
15 const char* ssid = "Fal's";
16 const char* password = "nopal2204";
17
18 // ----- TELEGRAM -----
19 String botToken = "8321482003:AAG7F7Lbhgitr1xwUdvVaE1D5hq45wEVp4Q";
20 String chatID = "897427425";
21
22 // ----- BLUETOOTH -----
23 uint8_t slaveAddress[] = {0x4C, 0xC3, 0x82, 0x0D, 0x5F, 0x1E};
24
25 // ----- PINS -----
26 #define HUSKY_SDA 32
27 #define HUSKY_SCL 33
28 #define TRIG_PIN 4
29 #define ECHO_PIN 5
30 #define MASTER_BUZZ_PIN 13
31 #define BUZZ_CH 0
32
33 // ----- SETTINGS -----
34 float FALL_THRESHOLD = 28.0;
35 unsigned long lastFallTime = 0;
36 const unsigned long FALL_COOLDOWN = 5000;
37 const int OBSTACLE_DIST_CM = 200;
38 unsigned long lastBeepTime = 0;
39 unsigned long lastBTCheck = 0;
40
41 // ----- FLAGS -----
42 volatile bool telegramBusy = false;
43 volatile bool fallTriggered = false;
44 String pendingMessage = "";
45
46 // ----- SEMAPHORE -----
47 SemaphoreHandle_t btMutex;
48
49 // ----- TELEGRAM TASK -----
50 void telegramTask(void * parameter) {
51     String msg = *((String*)parameter);
52     Serial.println("[TELEGRAM] Sending...");
53 }
```